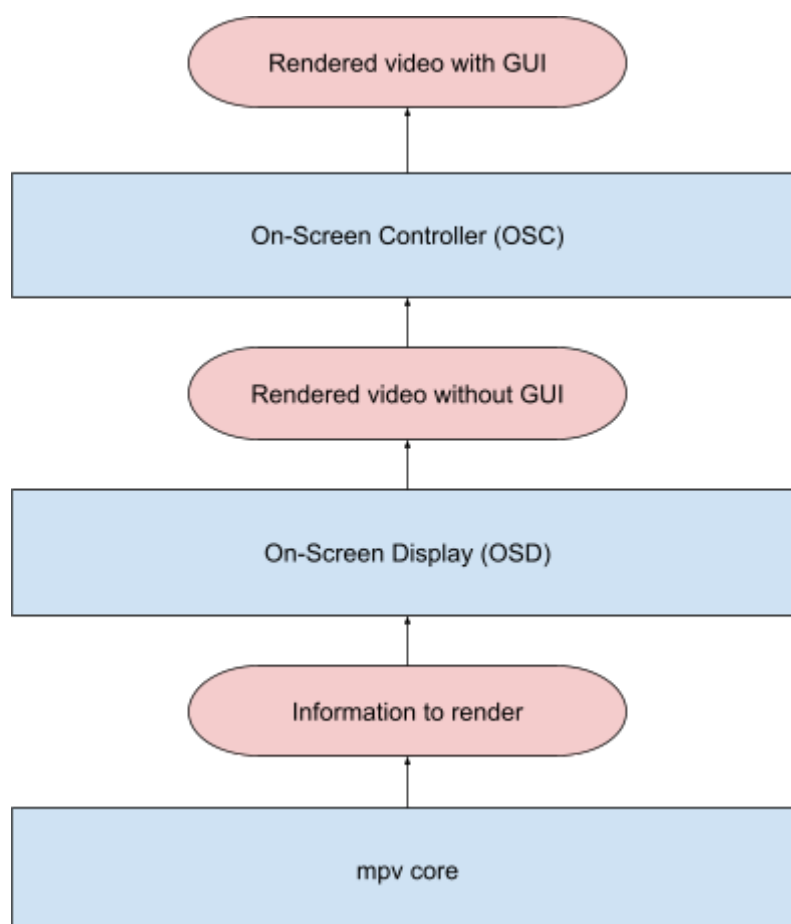


PIC1 Feature Proposal

In mpv, the OSD (On-Screen Display) is the graphics output layer that renders frames according to the instructions received from the mpv core code. Meanwhile, the OSC (On-Screen Controller) is a pseudo-GUI that only has the absolutely necessary features like a video seek bar, playlist controllers or buttons to switch between audio/captions tracks. Thus, the OSC provides a graphical way for the user to interact with the OSD, being built on top of this layer.

The OSC is currently written in Lua, which allows for easy customization of this pseudo-GUI by both users and developers. However, since the OSD is written in C, the interaction between these two layers is limited and it's harder to customize the display of the video fully.



We will implement an OSD layer in Lua so that third-party creators that wish to make their own OSC can do so more easily by also customizing more “basic” elements that belong to the OSD.

The elements that the OSD provides are simply visual elements like a volume bar and a seek bar that appear when the user scrolls horizontally or vertically, respectively, as well as the corresponding text (that clarifies what the user is changing). With an OSD layer written in Lua, it would be easy to change these bars' colors or the position of this text.

We'll be creating `player/lua/osd.lua` to implement the Lua-based OSD rendering layer, and will use the already existing file `player/lua.c`, that provides several functions to interact with the C layer and access its data, packaged in the mp module.

Thus, we will write Lua functions that implement similar features to the ones in the already coded OSD, for example:

- Listening for events such as play, seek, pause, just like the OSD in C;
- Use `mp.command()` to control video playback (this is a custom Lua command from the mp modules, in `player/lua.c`)
- Render text and progress bars with ASS formatting (the subtitle and text styling system used in mpv)

Since the OSC layer in `player/lua/osc.lua` also uses mp functions, we'll adopt a similar coding pattern.

Regarding the testing, besides manually verifying that the video is being correctly rendered and the OSC is correctly displaying on top of it, we'll create both unit and integration tests. The unit tests will ensure that values were not changed or lost between the two OSD implementations, while the integration tests will be similar to the ones that already exist, like `test/limpv_test_file_loading.c`, where we'll instead load the files with the Lua OSD rather than with C one.

We also plan on creating custom configuration files to personalize both the OSC and the OSD to confirm that it is indeed easier to do so now that both layers are written in Lua, like mentioned in the beginning of this proposal.

This feature was requested in the feature proposal back-log
(<https://github.com/mpv-player/mpv/issues/16991>)

Predicted LoC changes: 300-500 lines

Predicted unit tests: 400 lines

Predicted implementation effort: 18 hours

Predicted test & debugging effort: 10 hours